

MAA 5229 First-Year Exam, May 2019

Do exactly 4 problems. Work must be presented in a neat and logical fashion in order to receive credit. Do not leave any gaps. When a theorem is used in a proof it must be precisely stated.

1. Let $f_n : [a, b] \rightarrow \mathbb{R}$ be a uniformly bounded sequence of Riemann integrable functions. Prove that if $f_n \rightarrow f$ uniformly, then f is Riemann integrable.

2. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous. Suppose that

$$\int_0^1 f(x)x^{2019k} dx = 0$$

for all integers $k \geq 0$. Must f be identically 0? Prove, or give a counterexample.

3. Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a sequence of measurable functions and suppose there is a function $g \in L^1[0, 1]$ such that $|f_n| \leq g$ for all n . Consider the functions

$$F_n(t) = \int_0^t f_n(x) dx.$$

Prove i) each F_n is continuous on $[0, 1]$ and ii) there is a subsequence (F_{n_k}) converging uniformly on $[0, 1]$.

4. Let $(X, \mathcal{M})$ be a measurable space and $f_n : X \rightarrow \mathbb{R}$, $n = 1, 2, 3, \dots$ a sequence of measurable functions. Prove that each of the following subsets of X is measurable:
- a) $\{x | f_n(x) > 0 \text{ for infinitely many values of } n\}$
 - b) $\{x | \text{the sequence } (f_n(x)) \text{ is eventually monotone}\}$
 - c) $\{x | \lim_{n \rightarrow \infty} n f_n(x) = 0\}$

5. Let $f_1 \geq f_2 \geq f_3 \geq \dots$ be nonnegative measurable functions on a measure space $(X, \mathcal{M}, \mu)$, and put $f = \lim f_n$. Suppose that $\int f_k d\mu < \infty$ for some k . Prove that

$$\int f d\mu = \lim \int f_n d\mu.$$

Give a counterexample to show that the conclusion can fail if the finiteness hypothesis is dropped.